James A. Barron

PhD Candidate | Department of Physics, Engineering Physics & Astronomy,
Queen's University, Kingston, Canada
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Education

Ph.D. Candidate, Astrophysics Present
Queen's University, Kingston, Canada
Thesis: *Stellar Magnetism Through the Instability Strip*
Supervisor: Gregg A. Wade

Master of Science, Astrophysics
Queen's University, Kingston, Canada
Thesis: *A Systematic Study of the TESS Photometry of the Magnetic O-type Stars*, (ADS)
Supervisor: Gregg A. Wade

Bachelor of Science, Physics (Honours)
Queen's University, Kingston, Canada

Bachelor of Science, Specialization in Mathematics
University of Alberta, Edmonton, Canada

Refereed Publications

1. **Barron, J. A.**, Wade, G. A., Folsom, C. P. (2026), The rotational and magnetic properties of Polaris from long-term spectropolarimetric monitoring, ApJ, 1005, 1, 112, (ADS Link)
2. Biswas, A., Folsom, C.P., **Barron, J. A.**, Wade, G.A., Bellotti, S., Trigilio, C. (2026), Discovery of a rapidly evolving magnetic field in the M-dwarf YZ Cet and constraints on the magnetic field of its planet YZ Cet b, ApJ, 997, 2, 268 (ADS Link)
3. **Barron, J. A.**, Wade, G. A., Holgado, G., & Simón-Díaz, S. (2025), A Magnetic Field Detection in the Massive O-type Bright Giant 63 Oph, ApJ, 991, 79, (ADS Link)
4. Wade, G. A., Oksala, M., Neiner, C., Boucher, É., & **Barron, J. A.** (2025), Magnetic Field Monitoring of Four Massive A–F Supergiants, ApJL, 988, L38, (ADS Link)
5. Biswas, A., Das, B., **Barron, J. A.**, Wade, G. A., & Holgado, G. (2025), A Nonstop Aurora? The Intriguing Radio Emission from the Rapidly Rotating Magnetic Massive Star HR 5907, ApJ, 980, 2, 260, (ADS Link)
6. Frost, A.J., Sana, H., Mahy, L., Wade, G., **Barron, J.**, Le Bouquin, J.-B., Mérand, A., Schneider, F. R. N., Shenar, T., Barbá, R. H., Bowman, D. M., Fabry, M., Farhang, A., Marchant, P., Morrell, N. I., & Smoker, J. V. (2024), A magnetic massive star has experienced a stellar merger, Science, 384, 6692, 214-217, (ADS Link)
7. **Barron, J. A.**, Wade, G. A., Evans, N. R., Folsom, C. P., & Neilson, H. R. (2022), Finding magnetic north: an extraordinary magnetic field detection in Polaris and first results of a magnetic survey of classical Cepheids, MNRAS, 512, 4021-4030, (ADS Link)

Manuscripts to be Submitted

1. **Barron, J. A.**, Wade, G. A., Folsom, C. P., Kochukhov, O., First magnetic detection in a W Virginis Type II Cepheid: Modelling the shock-affected magnetic signature of κ Pavonis, in preparation for ApJ

Conference Proceedings

1. **Barron, J. A.**, Wade, G. A., Holgado, G., & Simón-Díaz, S., Discovery of a Magnetic Field in the O-type Bright Giant 63 Oph, IAUS, 402, In Press.
2. **Barron, J. A.**, Wade, G. A., Folsom, C. P., & Kochukhov, O. (2024), First results of a magnetic survey of classical Cepheids, IAUS, 361, 233-235, (ADS Link)
3. **Barron, J.**, Wade, G. A., Bowman, D. M., David-Uraz, A., Munoz, M. S., Pablo, H., & Simón-Díaz, S. (2020), MOBSTER: Identifying Candidate Magnetic O Stars through Rotational Modulation of TESS Photometry, Proceedings of the Polish Astronomical Society, 11, 226-235, (ADS Link)
4. David-Uraz, A., Neiner, C., Sikora, J., **Barron, J.**, Bowman, D. M., Cerrahoğlu, P., Cohen, D. H., Erba, C., Khalack, V., Kobzar, O., Kochukhov, O., Pablo, H., Petit, V., Shultz, M. E., ud-Doula, A., Wade, G. A. (2020), Magnetic OB[A] Stars with TESS: probing their Evolutionary and Rotational properties - The MOBSTER Collaboration, Proceedings of the conference Stars and their Variability Observed from Space, held in Vienna on August 19-23, 2019, (ADS Link)
5. David-Uraz, A.; Neiner, C.; Sikora, J.; **Barron, J.**; Bowman, D. M.; Cerrahoğlu, P.; Cohen, D. H.; Erba, C.; Kobzar, O.; Kochukhov, O.; Petit, V.; Shultz, M. E.; Ud-Doula, A.; Wade, G. A.; Mobster Collaboration (2020), MOBSTER: Establishing a Picture of Magnetic Massive Stars as a Population, Stellar Magnetism: A Workshop in Honour of the Career and Contributions of John D. Landstreet, held 8-11 July 2019 in London, Canada, (ADS Link)

Community Papers

1. **Barron, J. A.,** Wade, G. A., Bagnulo, S., Biswas, A., Bowman, D., David-Uraz, A., Erba, C., Eze, C. I., Folsom, C. P., Gutteridge, C., Khalack, V., Kritcka, J., Labadie-Bartz, J., Landstreet, J., Natan, T., Neiner, C., Oksala, M., Petit, V., Shultz, M., Marharyta, S., Stanley, P., Xia, Z. (2024), New capabilities for the Gemini Observatory: The case for high resolution optical spectropolarimetry, Gemini Observatory Community Brief Recommendation.

Invited Talks

Instituto de Astrofísica de Canarias, Tenerife, Spain — Oct 2022
Magnetic properties of classical pulsators

Contributed Talks

CASCA 2026: MIL-ion Futures of Astrophysics, Montréal, Québec — June 2026
The rotational and magnetic properties of Polaris: Challenges to standard stellar models

CFHT Users Meeting, Lac-à-l'Eau-Claire, Canada — May 2025
Stellar magnetism in the instability strip: New magnetic detections in classical and Type II Cepheids

MiMeS Workshop, Queen's University, Canada — May 2025
Follow-up of MiMeS magnetic O star candidates

MIAPbP Stellar Magnetic Fields Workshop, Garching, Germany — Oct 2023
Stellar magnetism through the instability strip

TASC6/KASC13 Workshop, KU Leuven, Belgium — Jul 2022
A systematic study of the TESS photometry of the magnetic O-type stars

MOBSTER-1 Virtual Conference Sep 2021
MOBSTER: Identifying candidate magnetic O Stars through rotational modulation of TESS photometry

Conference Posters

First Author

Barron, J. A., Wade, G. A., Holgado, G., & Simón-Díaz, S., Discovery of a Magnetic Field in the O-type Bright Giant 63 Oph, IAU Symposium 402: Massive Stars Across Redshifts in the Era of JWST and Large-Scale Surveys, Ensenada, Mexico, September 2025

Barron, J. A., Wade, G. A., Folsom, C. P., & Kochukhov, O., The magnetic fields of pulsating Cepheid variables, Magnetic Fields from Clouds to Stars (Bfields-2024), Tokyo, Japan, March 2024

Barron, J. A., Wade, G. A., Folsom, C. P., & Kochukhov, O., First results of a magnetic survey of classical Cepheids, RR Lyrae & Cepheid Stars Conference, La Palma, Spain, September 2022

Barron, J. A., Wade, G. A., Folsom, C. P., & Kochukhov, O., First results of a magnetic survey of classical Cepheids, IAU Symposium 361: Massive Stars Near and Far, Ballyconnell, Ireland, May 2022

Barron, J. A. O., & Wade, G. A. (2021), TESS photometry of the Of?p stars HD 148937 and LMC 164-2, Poster talk at OBA stars: Variability and Magnetic Fields, Virtual, St. Petersburg, (ADS Link)

Barron, J. A. O, Wade, G. A. (2021), Is the magnetic activity of Classical Cepheids modulated by pulsation? The case of eta Aquilae, MOBSTER-1 virtual conference: Stellar variability as a probe of magnetic fields in massive stars, July 2020, (ADS Link)

Co-Author

Biswas, A., Folsom, C. P., **Barron, J. A.**, & Wade, G. A. (2025), Zeeman Doppler Imaging of the Exoplanet Hosting M-dwarf System YZ Cet Reveals an Extreme Short-term Change in the Global Magnetic Field, Canada-France-Hawaii Telescope Users Meeting, Lac-à-l'Eau-Claire, Canada, May, 2025

Stacey, Erik; Wade, Gregg A.; **Barron, James**, (2021), Plaskett's Star: variability in the TESS photometric data, MOBSTER-1 Virtual Conference: Stellar variability as a probe of magnetic fields in massive stars, (ADS Link)

Wade, G. A., Woodcock, K., Lim, O., Bohlender, D., Monin, D., Sikora, J., **Barron, J.**, Shultz, M., Alecian, E., Neiner, C., & MiMeS Collaboration (2021), Two new magnetic B-type stars from the MiMeS survey, MOBSTER-1 Virtual Conference: Stellar variability as a probe of magnetic fields in massive stars, (ADS Link)

Funding and Awards

Natural Sciences and Engineering Research Council of Canada PGS-D 2023 – 2026

G. Neilson Whyte Graduate Fellowship 2025
Queen's University

Conference Travel Award 2022 – 2025
Queen's University, (3 awards)

Harold M. Cave Graduate Travel Scholarship 2023 – 2025
Queen's University, (3 awards)

R.S McLaughlin Fellowship 2022 – 2023
Queen's University

Carl Reinhardt Fellowship 2021 – 2022
Queen's University

Graduate Entrance Tuition Award 2019 – 2020
Queen's University

NSERC Undergraduate Research Award 2019
Royal Military College of Canada

Dean's Honour List 2018 – 2019
Queen's University

Harold M. Cave Undergraduate Travel Scholarship 2018
Queen's University

Service and Leadership

Observatory Volunteer, Queen's Observatory Sep 2023 – Present

Volunteer, Queen's Astronomy on Tap Sep 2023 – Present

Graduate Student Committee Representative, Canadian Astronomical Society Sep 2022 – Present

Graduate Oral Presentation and Poster Judge, CASCA 2026 June 2026

Proposal Reviewer, Canadian Time Allocation Committee, CFHT Apr 2026

Presenter, QUARG Journals Club Sep 2019 – Mar 2026

Category Judge, Frontenac, Lennox & Addington Science Fair Mar 2026

Adjudicator, Inquiry@Queen's Undergraduate Research Conference Mar 2026

Volunteer, Science Literacy Week, McDonald Institute & Queen's Observatory Oct 2025

Graduate Poster Grader, Canadian Astronomical Society Annual Meeting Jun 2025

Outreach Volunteer, Science Rendezvous Kingston May 2025

LOC Member and Session Chair, MiMeS Workshop, Queen's University May 2025

Outreach Volunteer, Science Rendezvous Kingston May 2024

Volunteer, Queen's Eclipse Ambassador Sep 2023 – Apr 2024

Graduate Steering Representative, Queen's Graduate Physics Society Sep 2022 – Aug 2023

Adjudicator, Inquiry@Queen's Undergraduate Research Conference Mar 2022

Category Judge, Frontenac, Lennox & Addington Science Fair Mar 2022

LOC Member and Session Chair, MOBSTER-2 Virtual Conference Nov 2021

Science Fair Judge, Canada-Wide Science Fair May 2021

Category Judge, Frontenac, Lennox & Addington Science Fair Mar 2021

Let's Talk Astrophysics Volunteer, Let's Talk Science, Queen's University Feb 2021

Outreach Presentation, Greenview Elementary School, Edmonton, Canada Dec 2020

Conference Volunteer, The Physics of Galaxy Scaling Relations, Queen's University Jul 2018

Research Mentorship

Undergraduate Research (NSERC USRA), Royal Military College of Canada May 2026 - Present
Development of software analysis pipeline for Gaia DR4

Undergraduate Research Thesis, Queen's University Sep 2025 – Apr 2026
Inferring the atmospheric velocity gradients of classical Cepheids

Undergraduate Research (NSERC USRA), Royal Military College of Canada — May 2024 – Aug 2024
A first BRITE, SMEI and TESS analysis of the suspected merger product θ-Scorpii
Presented at the 2024 BRITE Side of Stars Conference, Vienna, Austria, (ADS Link)

Undergraduate Research (NSERC USRA), Royal Military College of Canada — May 2023 – Aug 2023
Searching for slowly rotating magnetic massive stars with Gaia

Undergraduate Research Project, Royal Military College of Canada — May 2021 – Aug 2021
Analysis of time-series photometry of a B-type supergiant

Teaching Experience (Queen's University)

Teaching Assistant | Introduction to Physics II (PHYS 116), 1 semester — Jan 2026 – Apr 2026

Teaching Assistant | Experimentation and Design (ASPC 102), 1 semester — Sep 2025 – Dec 2025

Teaching Assistant | The Solar System (ASTR 101), 4 semesters — Sep 2020 – Apr 2024

Teaching Assistant | Introduction to Physics Tutorial, (PHYS 115), 1 semester — Sep 2023 – Dec 2023

Teaching Assistant | Stellar Structure and Evolution (PHYS 435), 2 semesters — Sep 2022 – Dec 2023

Teaching Assistant | Physical Processes in Astrophysics (PHYS 315), 1 semester — Sep 2022 – Dec 2022

Teaching Assistant | Stars, Galaxies, and the Universe (ASTR 102), 3 semesters — Sep 2020 – Dec 2022

Teaching Assistant | Introductory Physics Laboratory (PHYS 117), 3 semesters — Sep 2019 – Apr 2022

Professional Development

Research Project Management Training — Jan 2026
Canadian Association for Graduate Studies

Science Leadership and Management — Dec 2022
Queen's University (semester course)

Introduction to Astronomical Instrumentation — Jul 2021
Dunlap Institute Summer School, University of Toronto

Summer School on High Performance and Technical Computing — Aug 2019
Centre for Advanced Computing, Queen's University

CRAQ Summer School on Stellar Astrophysics — Jun 2019
Centre for Research in Astrophysics of Quebec, McGill University

Observing Programs

Principal Investigator

Finding Magnetic North: A first magnetic map of the classical Cepheid Polaris — 2022 – 2025
ESPaDOnS/Canada-France-Hawaii Telescope, 25 hours (6 semesters)

Characterization of new FS CMa candidates with GHOST: An overlooked channel for post-merger objects — 2025
GHOST/Gemini South, 2 hours (GS-2025A-FT-219)

Concluding the first systematic magnetic study of classical Cepheids: probing the low and high mass regions of the instability strip — 2024
ESPaDOnS/Canada-France-Hawaii Telescope, 13.5 hours (24BC27)

A first systematic study of the magnetic fields of classical Cepheid stars 2022 – 2023
HARPS/ESO 3.6-metre Telescope, 5 nights (108.22E7.001, 111.24ZW.001)
Three nights of observing experience as visiting astronomer at La Silla observatory.

A first systematic study of the magnetic fields of classical Cepheid stars 2020 – 2022
ESPaDOnS/Canada-France-Hawaii Telescope, 53.6 hours (4 semesters)

Follow-up observations of two magnetic O star candidates identified in the MiMeS survey 2022
ESPaDOnS/Canada-France-Hawaii Telescope, 7.2 hours (22AC31)

Confirmation of Magnetic O Star Candidates Identified from Rotational Modulation of TESS Photometry 2020
ESPaDOnS/Canada-France-Hawaii Telescope, 9.6 hours (20BC31)

Co-Investigator

Spectropolarimetric Followup of Indirect Magnetospheric Signatures from Multi Wavelength Observation of a Rapidly Rotating Massive Star 2025
ESPaDOnS/Canada-France-Hawaii Telescope, 2 hours (25BC23)

Monitoring the magnetic fields of the A-type supergiants 13 Mon and eta Leo 2025
ESPaDOnS/Canada-France-Hawaii Telescope, 5.4 hours (25BF11)

Detailed Study of the Magnetic O-star Binary HD 148937: The First Ever Magnetic O-star System Detected at Sub-GHz Frequencies 2025
Giant Microwave Radio Telescope, 35 hours (49_067)

Follow-up uGMRT observation of the most rapidly-rotating, non-degenerate, magnetic massive star: HD 142184 2024
Giant Microwave Radio Telescope, 14 hours (46_065)

Constraining the magnetic field of YZ Cet b through star-planet interactions 2024
SPIRou/Canada-France-Hawaii Telescope, 2.6 hours (24BC28)

The unusual case of the double star HD 155273: a potential hidden magnetic B-type binary 2022
ESPaDOnS/Canada-France-Hawaii Telescope, 4.6 hours (22AC33)

Magnetism in the upper Hertzsprung-Russell Diagram: using TESS to characterize the distinct sub-population of magnetic OBA stars and hunt for new magnetic candidates 2022
NASA Astrophysics Data Analysis Program (22-DA_ADAP22-0070)